

Moving Average

Types of moving average

- ① SMA (Simple moving avg)
- ② CMA (Cumulative moving avg)
- ③ EMA Or EWMA (Exponential moving avg)

Simple moving avg

$$\text{Average} = [10, 12, 15, 13, 11]$$

$$\text{Average} = \frac{\text{add all the val}}{\text{No. of values}}$$

$$\text{Average} = \frac{10+12+15+13+11}{5} = \boxed{12.2}$$

time sales value
↓
|

Why we calculate MA ?

SMOOTHING of the data

↓
{Time Series Data}

→ Moving ⇒ move over the time axis in specific window time

- ① Window size
- ② Average calculation

Example

① window = 3

② avg

	D1	10	NAN NAN
	D2	12	
	D3	15	→ 12.33
	D4	13	→ 13.33
	D5	14	→ 14
	D6	16	→ 14.33
	D7	17	→ 15.66

$$1^{st} \text{ avg} = \frac{D_1 + D_2 + D_3}{3} = \frac{10 + 12 + 15}{3} = 12.33$$

$$2^{nd} \text{ avg} = \frac{D_2 + D_3 + D_4}{3} = \frac{12 + 15 + 13}{3} = 13.33$$

$$3^{rd} = \frac{D_3 + D_4 + D_5}{3} = \frac{15 + 13 + 14}{3} = 14$$

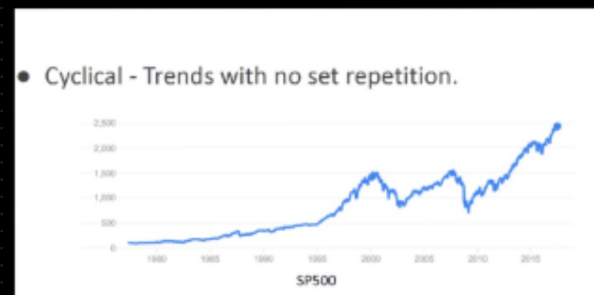
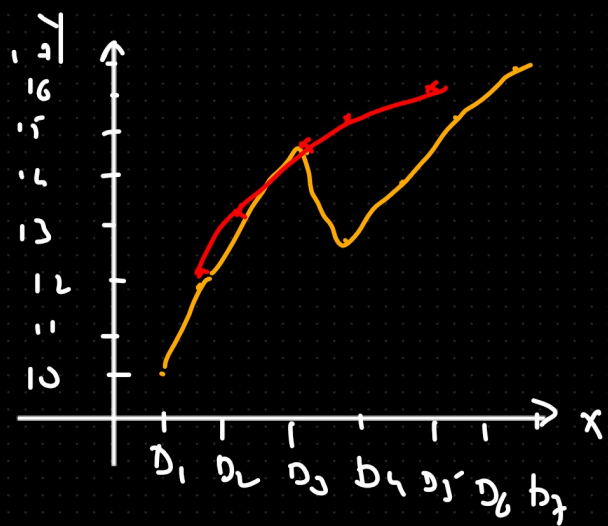
$$4^{th} = \frac{D_4 + D_5 + D_6}{3} = \frac{13 + 14 + 16}{3} = 14.33$$

$$5^{th} = \frac{D_5 + D_6 + D_7}{3} = \frac{14 + 16 + 17}{3} = 15.66$$

SMOOTHING ⇒ To remove all the effect from the data.

We can perform smoothing with the help of MA

- ① Pattern recognition from the data
- ② You can analysis the trend of the data
- ③ You can reduce the effect of outlier
- ④ Enhancing the visualization



CMA $\Rightarrow$ find out the avg. of all the data point up to the given time stamp.

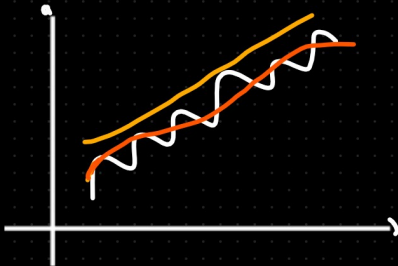
CMA

$- D_1$	10	$D_1 = 10$
$- D_2$	12	$D_1 + D_2 / 2 = 10 + 12 / 2 =$
$- D_3$	15	$10 + 12 + 15 / 3 =$
$\rightarrow D_4$	14	$10 + 12 + 15 + 14 / 4 =$
D_5	16	$10 + 12 + 15 + 14 + 16 / 5 =$
D_6	17	$10 + 12 + 15 + 14 + 16 + 17 / 6 =$
D_7	18	$10 + 12 + 15 + 14 + 16 + 17 + 18 / 7 =$

① We use CMA for the long time period.

② It will give you exponential trend

SMOOTHING



EMA or EWMA $\Rightarrow$ In EMA we give more weightage to the recent data point or give more weightage to the recent time stamp.

$\boxed{D_1}$ $\rightarrow$ low
 D_2
 D_3
 D_4
 $\boxed{D_5}$ $\rightarrow$ more w

$$\boxed{V_t = \beta V_{t-1} + (1-\beta) \theta_t}$$

V_t = EMA at time t

$\beta \Rightarrow 0 < \beta < 1 \Rightarrow \boxed{0.9}$

V_{t-1} = EMA at previous time stamp

θ_t = Data at t time stamp

$V_0 \rightarrow$	D_1	25	= 0
$V_1 \rightarrow$	D_2	13	= 1.3
$V_2 \rightarrow$	D_3	17	= 2.87
$V_3 \rightarrow$	D_4	31	
$V_4 \rightarrow$	D_5	43	

$$\boxed{V_0 = 0} \quad \text{or} \quad \boxed{V_0 = 25}$$

$$V_1 = \beta \times V_0 + (1-\beta) \theta_1$$

$$V_1 = 0.9 \times 0 + (1-0.9) 13$$

$$V_1 = 0 + 0.1 \times 13 = 1.3$$

$$V_2 = \beta \times V_1 + (1-\beta) \theta_2$$

$$V_2 = 0.9 \times 1.3 + 0.1 \times 17$$

$$V_2 = 1.17 + 1.7$$

$$V_2 = 2.87$$

